

THIS IS A GRADED ASSIGNMENT! Answers must be entered in MOODLE before the due date/time.

PROBLEM #1.

You have sampled a population in which you know that the percentage of the homozygous recessive genotype (aa) is 36%. Using that 36%, calculate the following. **NOTE: All of your answers should be expressed as a percentage using one decimal place. For example, a HW value of 0.183 would be 18.3% as an answer. The value 0.183 is a relative frequency, whereas 18.3% is a percentage. While they seem the same, MOODLE will only accept a certain format for answers as it deems correct for an answer. ALL answers MUST have one decimal place.**

- A. The percentage of the "aa" genotype. Answer: **36.0%**, as given in the problem itself. This would correspond to a relative frequency of 0.36, but the answers should be expressed as a percentage and that is why you would enter 36.0 as your answer in MOODLE (do NOT include the "%" sign). This same rationale applies for the rest of the quiz in terms of how you enter your answers in MOODLE, due to the fact that MOODLE will only accept an answer as fully correct if it is entered in the format that it is expecting. Anything other than that, it will mark your answer as incorrect. You MUST INCLUDE one decimal place in your answer. For example, MOODLE will mark the following answers incorrect "36, 36%, 0.36, 36 percent"; only the answer "36.0" would be marked correct because it has one decimal place and is the correct value. The same approach applies for the rest of the answers to be input in MOODLE.
- B. The percentage of the "a" allele. Answer: The frequency of aa is 36%, which means that $q^2 = 0.36$, by definition. If $q^2 = 0.36$, then $q = 0.6$, again by definition. Since q equals the frequency of the a allele, then the frequency is **60.0%**.
- C. The percentage of the "A" allele. Answer: Since $q = 0.6$, and $p + q = 1$, then $p = 0.4$; the frequency of A is by definition equal to p, so the answer is **40.0%**.
- D. The percentages of the genotypes "AA" and "Aa." Answer: The frequency of AA is equal to p^2 , and the frequency of Aa is equal to $2pq$. So, using the information above, the frequency of AA is **16.0%** (i.e. p^2 is $0.4 \times 0.4 = 0.16$) and Aa is **48.0%** ($2pq = 2 \times 0.4 \times 0.6 = 0.48$).
- E. The percentages of the two possible phenotypes if "A" is completely dominant over "a." Answers: Because "A" is totally dominate over "a", the dominant phenotype will show if either the homozygous "AA" or heterozygous "Aa" genotypes occur. The recessive phenotype is controlled by the homozygous aa genotype. Therefore, the frequency of the dominant phenotype equals the sum of the frequencies of AA and Aa, and the recessive phenotype is simply the frequency of aa. Therefore, the dominant phenotype frequency is **64.0%** and, in the first part of this question above, you have already shown that the recessive phenotype frequency is **36.0%**.

PROBLEM #2.

Cystic fibrosis is a recessive condition that affects about 1 in 2,500 babies in the Caucasian population of the United States. Calculate the following.

- A. The percentage of the recessive allele in the population. Answer: We know from the above information that q^2 is 1/2,500 or 0.0004. Therefore, q is the square root, or 0.02. That is the answer to our first question: the frequency of the cystic fibrosis (recessive) allele in the population is 0.02 (or **2.0%**).
- B. The percentage of the dominant allele in the population. Answer: The frequency of the dominant (normal) allele in the population (p) is simply $1 - 0.02 = 0.98$ (or **98.0%**).
- C. The percentage of heterozygous individuals (carriers) in the population. Answer: Since $2pq$ equals the frequency of heterozygotes or carriers, then the equation will be as follows: $2pq = (2)(.98)(.02) = 0.0392$ (or **3.9%**) or ~1 in 25 are carriers.

PROBLEM #3.

In a given population, only the "A" and "B" alleles are present in the ABO system; there are no individuals with type "O" blood or with O alleles in this particular population. If 200 people have type A blood, 50 have type AB blood, and 50 have type B blood, what are the allelic frequencies (expressed as a percentage) of this population (i.e., what are p and q)?

Answer: To calculate the allele frequencies for A and B, we need to remember that the individuals with type A blood are homozygous AA, individuals with type AB blood are heterozygous AB, and individuals with type B blood are homozygous BB. The frequency of A equals the following: $2 \times (\text{number of AA}) + (\text{number of AB})$ divided by $2 \times (\text{total number of individuals})$. Thus $2 \times (200) + (50)$ divided by $2 \times (200 + 50 + 50)$. This is $450/600 = 0.750$ (or **75.0%**) = p (or the "A allele"). Since q (or the "B allele") is simply $1 - p$, then $q = 1 - 0.750$ or 0.250 (or **25.0%**).

1A: The percentage of the "aa" genotype	36.0	1E: The percentage of the recessive phenotype (or "aa")	36.0
1B: The percentage of the "a" allele	60.0	2A: The percentage of the recessive allele in the population	2.0
1C: The percentage of the "A" allele	40.0	2B: The percentage of the dominant allele in the population	98.0
1D: The percentage of the genotype "AA" (or homozygous dominant)	16.0	2C: The percentage of heterozygous individuals (carriers) in the population	3.9
1D: The percentage of the genotype "Aa" (or heterozygous)	48.0	3A: The percentage of the "A" allele in the population (or "p" for the HW formula)	75.0
1E: The percentage of the dominant phenotype (or "A_")	64.0	3B: The percentage of the "B" allele in the population (or "q" for the HW formula)	25.0